



**Petroleum Engineering Department
College of Engineering
Almaaqal University
To
Dr. Khalaf Gad Salem
By**

- SAJJAD KHALDOON HANO
- ABDUL HUSSEIN JABBAR NASER
- SAJJAD AHMED ABDULSAMED
- HUSSAIN LAZIM QASIM
- FAHAD KHALID SAAD
- AHMED FAHAD SALEH
- MAHER MOHAMMED QASIM
- SAJJAD ABBAS ALI
- ABBAS JUMAA ABDEL NABI

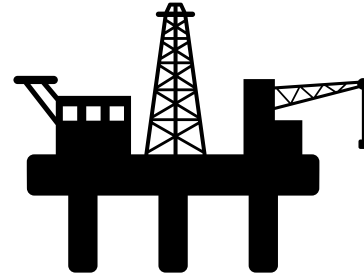
Field Development Plans

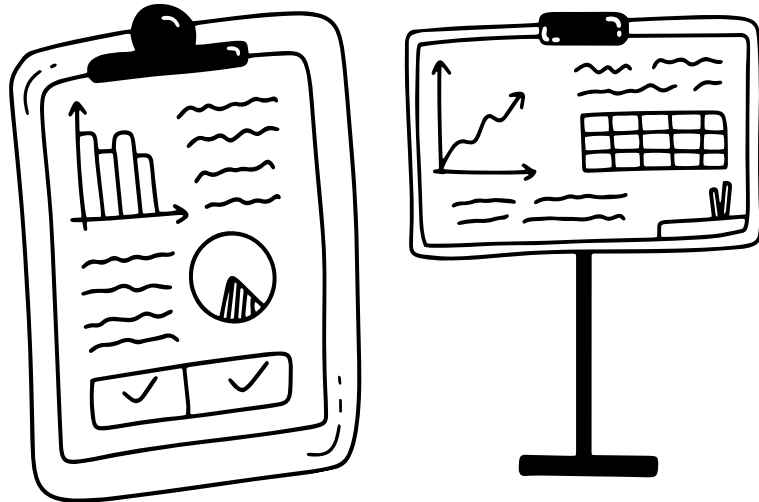
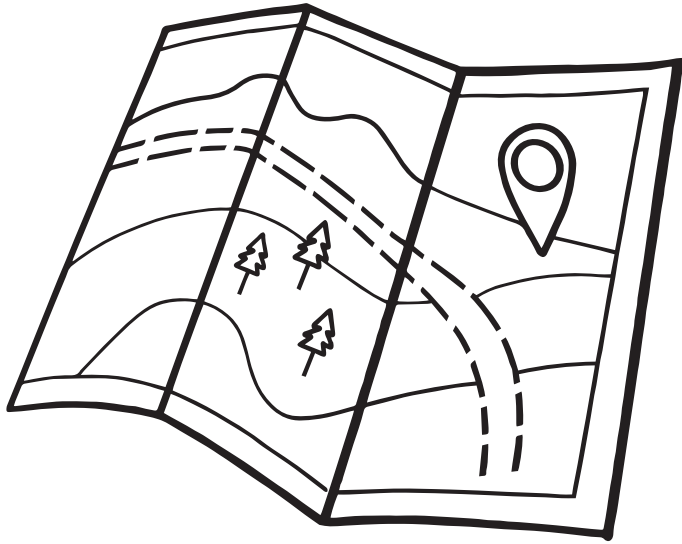
**A COMPREHENSIVE STRATEGY OR TRANSFORMING DISCOVERED OIL AND GAS
FIELDS IN TO PRODUCING ASSETS ,FROM EXPLORATION THROUGH ABANDONMENT**



1. Research Contents

- 2. Field Development Plan Define**
- 3. Initial Planning and Feasibility Studies**
- 4. Design and Engineering Phases**
- 5. Drilling and Well Completion**
- 6. Production Facilities and Infrastructure**
- 7. Field Development and Optimisation**
- 8. Well Abandonment and Decommissioning**
- 9. Cost Considerations and Economic Indicators**
- 10. Conclusion: From Discovery to Abandonment**
- 11. Sources**





What is a Field Development Plan?

An FDP is a blue print integrating technical ,economic, and environmental

considerations to extract hydrocarbons efficiently, safely, and profitably. It guides investment decisions, regulatory approvals, and operational execution.

Define Well Strategy

Optimal number and location of wells to maximise field value.

Plan Infrastructure

Production facilities and surface infrastructure design.

Recovery Optimisation

Strategies to maximise hydrocarbon extraction and profitability.

Initial Planning and Feasibility Studies

The foundation phase determines whether a discovered field can be developed economically and technically.

01

Data Acquisition

Integrate geological, geophysical, and petrophysical data to assess reservoir size and quality.

02

Reserves Estimation

Classify hydrocarbons using volumetric and simulation methods.

03

Development Scenarios

Evaluate production strategies and well configurations.

04

Economic Analysis

Cost-benefit analysis and sensitivity studies under varying market conditions.

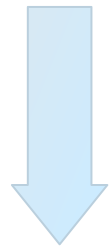
05

Risk Assessment

Identify technical, economic, and environmental risks with mitigation strategies.

Design and Engineering Phases

Detailed engineering translates the selected development concept into actionable design documents and specifications.



Conceptual Design

Define field layout, well locations, and production system configuration.



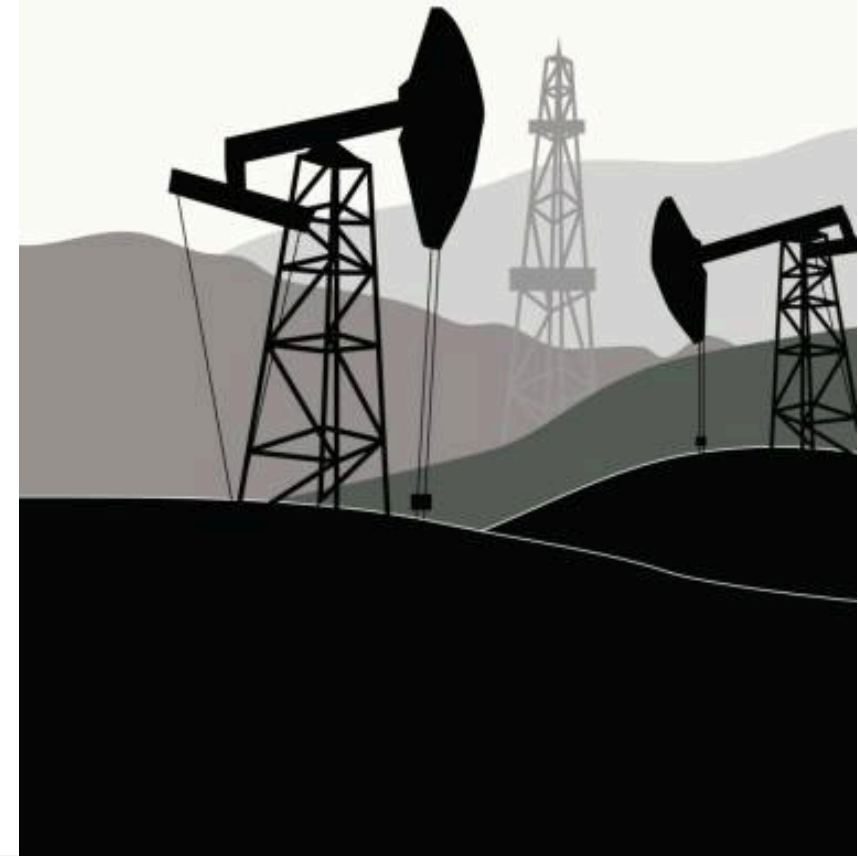
FEED Stage

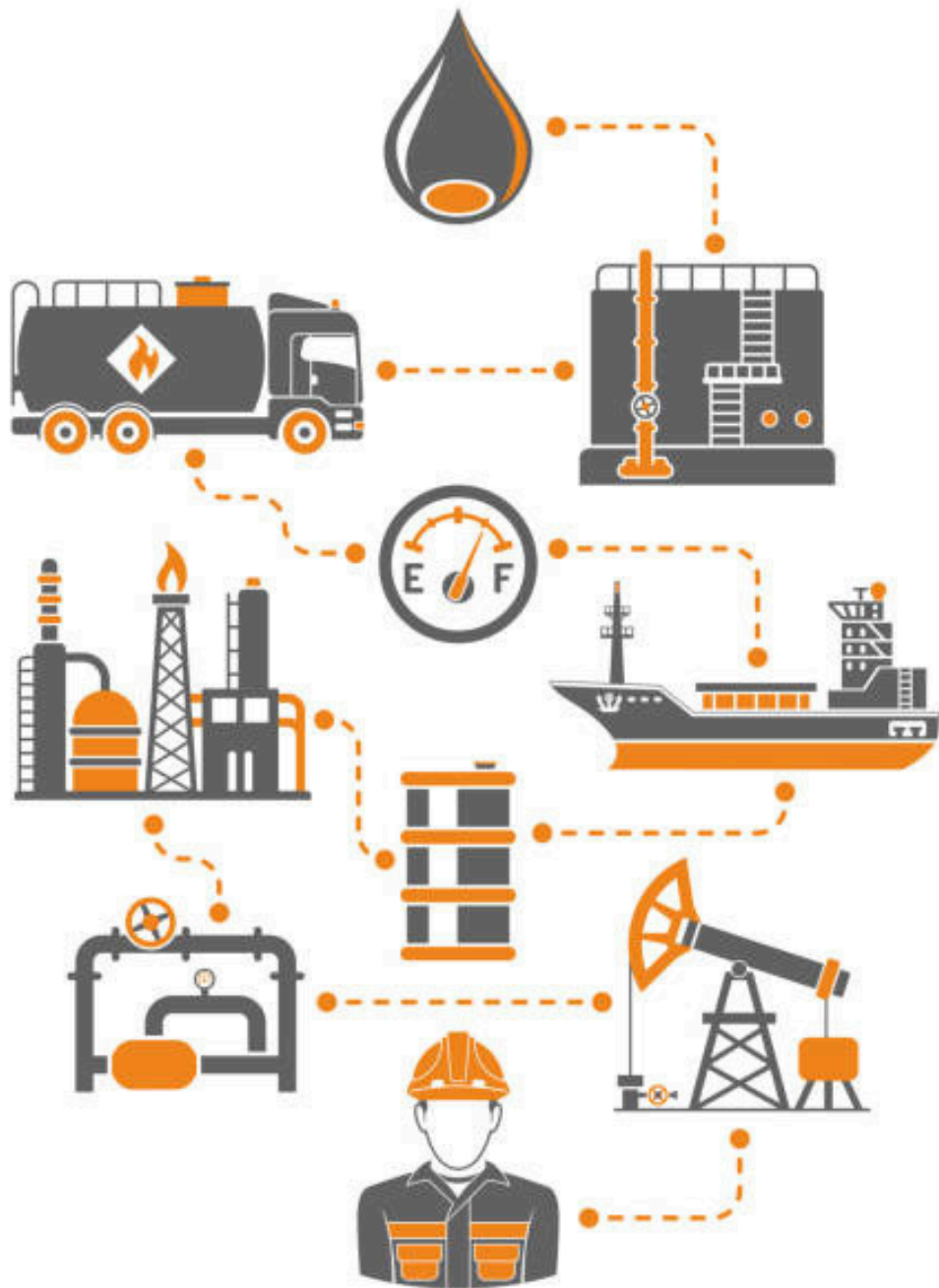
Refine parameters, specify equipment, and prepare accurate cost estimates with PFDs and P&IDs.



Detailed Engineering

Finalise construction drawings, specifications, and procurement documents for all systems.





Drilling and Well Completion

Critical phase involving direct interaction with the reservoir to achieve production targets safely and efficiently.

Drilling Operations

- Well trajectory and casing programme design
- Directional and horizontal drilling for enhanced reservoir contact
- Managed pressure drilling in challenging formations

Well Completion

- Casing and cementing for zone isolation
- Perforation and stimulation techniques
- Artificial lift systems installation

Production Facilities and Infrastructure

Processing and transport systems convert raw hydrocarbons into export-ready products.



Separation Units

Separate oil, gas, and water phases.



Compression Systems

Prepare gas for transport and export.



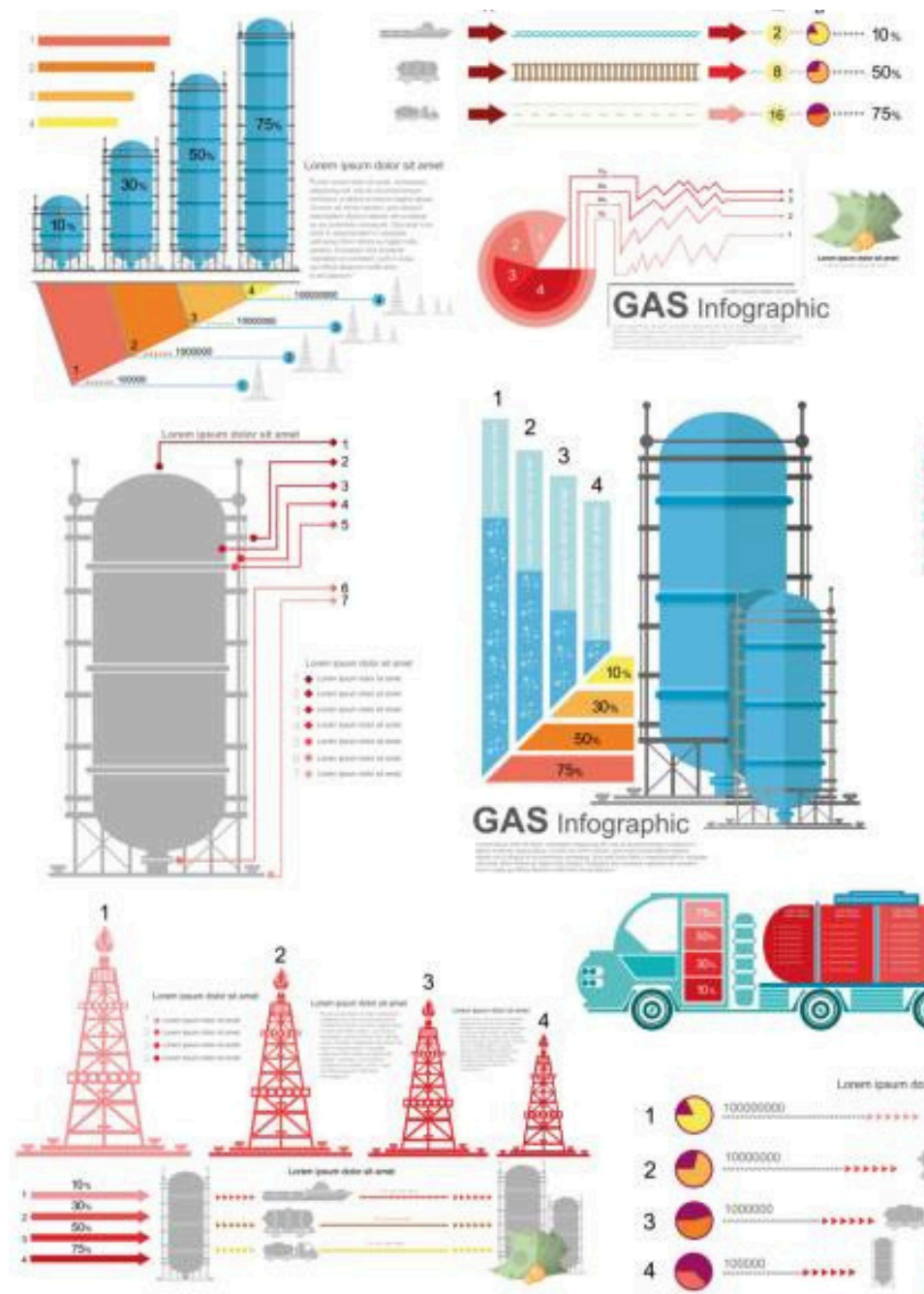
Storage and Export

Tanks, pipelines, and terminals for product delivery.



Environmental Control

Manage flaring, produced water, and emissions.



Field Development and Optimisation

Continuous monitoring and optimisation maintain performance and extend field life through production.





Well Abandonment and Decommissioning

Safe and environmentally responsible closure ensures compliance and environmental protection at end of field life.

1 Well Plugging

Permanent barriers isolate hydrocarbon-bearing zones.

2 Facility Decommissioning

Dismantling of platforms, pipelines, and processing units.

3 Site Restoration

Rehabilitation to meet environmental standards and regulatory requirements.

Cost Considerations and Economic Indicators

Cost management is central to every phase. Economic indicators evaluate profitability and guide investment decisions.

CAPEX

Capital Expenditure
Drilling, equipment, and construction costs.

OPEX

Operating Expenditure
Routine operations, maintenance, and manpower.

NPV

Net Present Value
Primary profitability indicator.

IRR

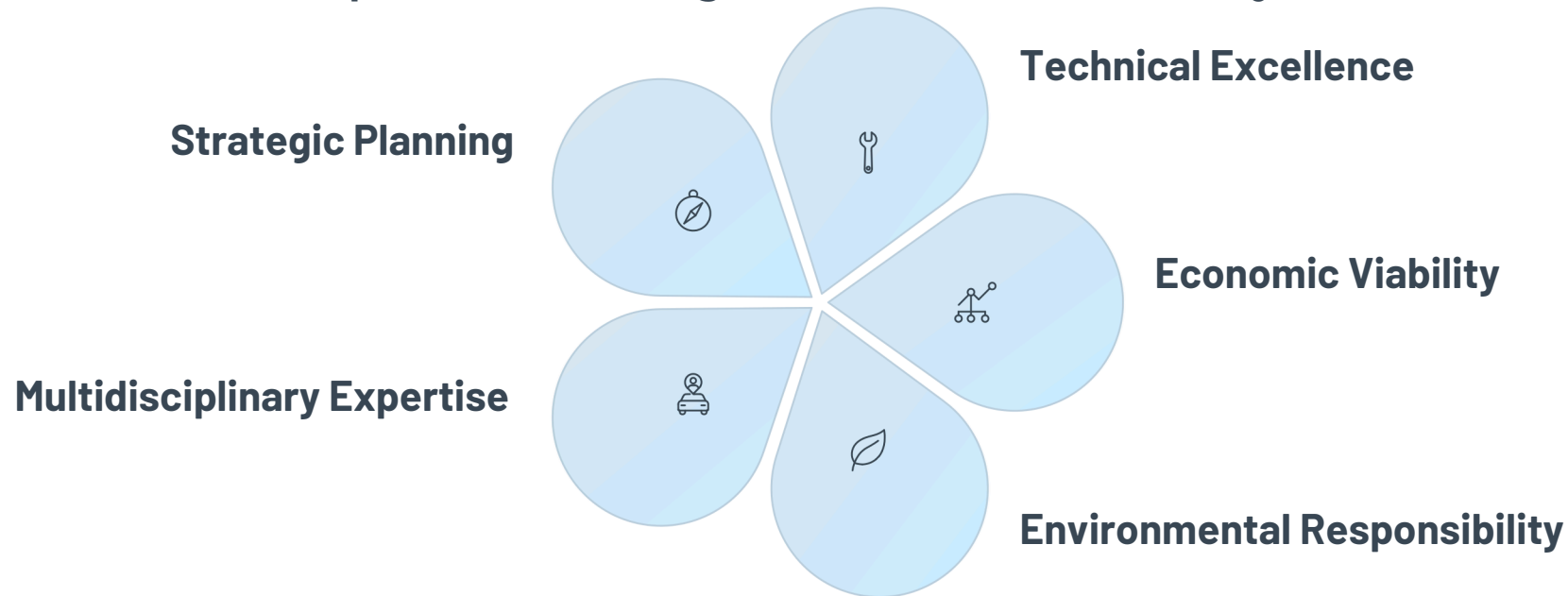
Internal Rate of Return
Investment return measurement.

Sensitivity and uncertainty analyses account for fluctuating oil prices, technical risks, and regulatory changes.

Conclusion: From Discovery to Abandonment

An effective Field Development Plan is a dynamic ,multi disciplinary roadmap transforming discovered hydrocarbon

accumulations into producing assets. Success relies on accurate data ,robust engineering, cost control , and continuous optimisation throughout the field's entire life cycle.



Sources

1. The Oil & Gas Authority (UK) – Revised Field Development Plan (FDP) Guidance

- A guidance document from the UK regulator covering technical, emissions and economic assessments.
- Useful to reference regulatory and approval-process aspects of FDPs.

2. The Commonwealth Secretariat – Field Development Plans: A Handbook for Government Officials

- A handbook designed to support government reviewers in assessing FDPs.
- Good for understanding how governments review, approve and oversee FDPs.

3. American Petroleum Institute (API) – Standards Overview

- API develops technical standards for oil & gas industry operations.
- Provides engineering/operational context that is useful for design/engineering phases.

4. ScienceDirect / Engineering Topics – Field Development: An Overview

- Summary article describing the phases of field development (exploration → development → operation → decommissioning).
- Useful for high-level lifecycle structure of a development project.

5. Academic Case Study – “Field developmental plan analysis: a case study of ‘x’”

- A published paper analyzing an FDP, integrating geology, reservoir engineering and economics.
- Good for detailed methodology, simulation and analysis aspects.

6. Specific Field Study – Integrated Field Development Plans of Akkas Gas Field

- A research article on field development planning process and economic analysis in a case field.

7. Guidelines for Upstream Facilities – Nigerian Upstream Petroleum Regulatory Commission (NUPRC) – Guidelines for the Design, Construction and Operation of Oil & Gas Production Facilities (Ver. 5, Nov 2023)

- Sets baseline requirements for facilities. Useful for infrastructure/production-facilities section.